

1. Summary

This report documents the implementation and experimental study of a decision-tree classifier on the Adult income dataset. The notebook implements several impurity and split criteria like: Gini index, entropy/information gain, misclassification error, and gain ratio, together with support for continuous and categorical attributes, missing-value handling, tree-depth control, and custom evaluation metrics. Five experimental tasks investigate criterion comparison, root-node split behavior, information-gain bias, overfitting under increasing tree complexity and label noise, and final hyperparameter selection.

The main observations are that (i) at a fixed maximum depth of 5, misclassification-error reduction gives the highest validation accuracy among the four criteria tested (84.74%); (ii) *relationship* is the strongest root feature under both Gini reduction and information gain, while gain ratio selects a *capital-gain* threshold; (iii) a synthetic *UniqueID* feature exposes the tendency of plain information gain to prefer high-cardinality identifiers; (iv) increasing tree depth steadily raises training accuracy while validation accuracy falls, demonstrating overfitting; and (v) the 10% label-noise experiment makes this degradation more severe.

2. Dataset and Preprocessing

The notebook loads `data/adult.data` and assigns 15 columns: 14 input attributes and the binary income target. The target is converted to 1 for records whose income contains “>50K” and to 0 otherwise.

Item	Implementation / Recorded Value
Dataset file	<code>data/adult.data</code>
Rows in original dataset	32,561
Input features	14
Target	income
Target encoding	>50K → 1; otherwise → 0
Train/validation split	80% / 20%
Training samples	26,048
Validation samples	6,513
Random seed	1332

Feature types and preprocessing behavior

The data contain both numerical and categorical variables. The notebook explicitly identifies categorical columns for exploratory distribution checks, but the custom tree retains categorical values directly rather than one-hot encoding them. During tree fitting, numeric missing values are imputed with the median and categorical missing values with the mode. Continuous splits are generated at midpoints between consecutive sorted unique values; categorical variables are handled by multiway branches.

- Numerical attributes include *age*, *fnlwgt*, *education_num*, *capital_gain*, *capital_loss*, and *hours_per_week*.
- Categorical attributes include *workclass*, *education*, *marital_status*, *occupation*, *relationship*, *race*, *sex*, and *native_country*.

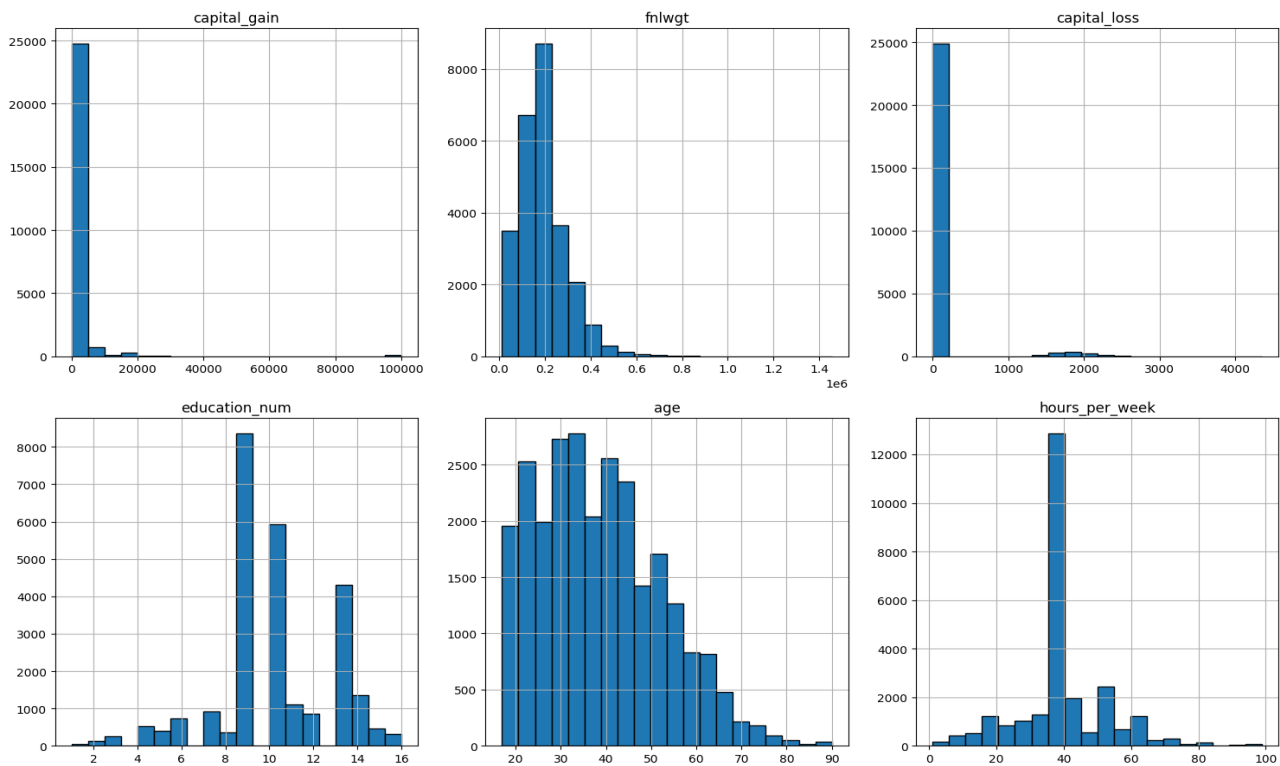


Figure 1. Recorded training-set histograms for numerical attributes. The distributions of `capital_gain` and `capital_loss` are highly concentrated near zero, while `fnlwgt` is right-skewed.

3. Methodology

The classifier is implemented from scratch. A Node stores the selected feature, threshold (for continuous splits), branch children, leaf value, and a default majority-class value. The `DecisionTreeClassifier` recursively searches for the split with the highest criterion score until a stopping condition is reached.

3.1 Impurity and split measures

For a node with class probabilities $p_1, p_2, \dots$, the notebook implements the following measures:

Measure	Formula / Interpretation
Entropy	$H(S) = - \sum p_i \log_2(p_i)$
Gini index	$G(S) = 1 - \sum p_i^2$
Misclassification error	$E(S) = 1 - \max_i p_i$
Information gain	$IG = H(\text{parent}) - \sum (S_j / S) H(S_j)$
Split information	$SI = - \sum (S_j / S) \log_2(S_j / S)$
Gain ratio	$GR = IG / SI$

For entropy, Gini, and misclassification error, the tree maximizes the reduction from parent impurity to weighted child impurity. Gain ratio is computed as information gain divided by split information, which normalizes information gain by the intrinsic fragmentation of the split.

3.2 Split generation

- **Continuous features:** candidate thresholds are the midpoints between consecutive unique sorted values.
- **Categorical features:** each distinct category creates a branch in the multiway split used by the main tree implementation.

- **Missing values:** *median* for numeric columns and *mode* for categorical columns during split search; a stored default majority class is used during traversal when a value is missing or an unseen branch is encountered.

4. Task 1 — Criterion Comparison

Each criterion was evaluated with maximum tree depth fixed at 5. The model was trained on the 26,048 training records and evaluated on the 6,513-record validation set. The resulting validation accuracies are recorded below.

Criterion	Maximum depth	Validation accuracy
Gini	5	84.49%
Entropy / Information Gain	5	84.05%
Misclassification Error	5	84.74%
Gain Ratio	5	82.14%

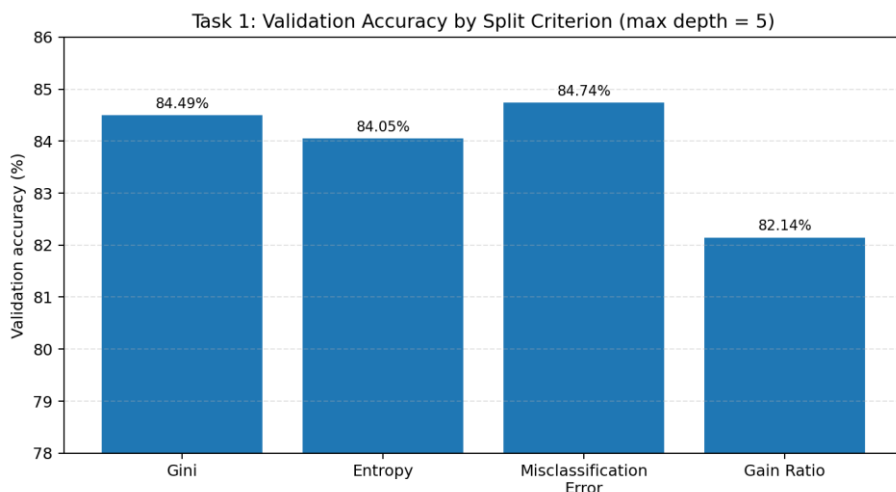


Figure 2. Task 1 validation accuracy for the four split criteria at maximum depth 5.

Analysis: misclassification-error reduction is the best performer in this fixed-depth comparison, followed closely by Gini and entropy. Gain ratio trails the other three criteria by more than two percentage points on this experiment.

5. Task 2 — Root Split Analysis and Manual Calculations

5.1 Best root split by criterion

Criterion	Selected root split	Score
Gini reduction	relationship (6-category multiway split)	0.0742
Information gain	relationship (6-category multiway split)	0.1643
Gain ratio	capital_gain $\leq$ 7073.5000	0.3420
Misclassification-error reduction	capital_gain $\leq$ 5119.0000	0.0431

5.2 Multiway versus binary partitioning of marital_status

For *marital_status*, the full seven-branch multiway split achieves *information gain* 0.1553 and *gain ratio* 0.0850. Among the binary partitions enumerated in the notebook, the strongest information-gain result is obtained by grouping {Married-AF-spouse, Married-civ-spouse} versus the remaining categories, with information gain 0.1514 and gain ratio 0.1521. The binary search illustrates that a carefully selected partition can approach the predictive separation of the multiway split while also changing the normalization used by gain ratio.

5.3 Worked split calculation: age ≤ 30

The notebook explicitly calculates all four quantities for the binary split *age* ≤ 30 versus *age* > 30.

Node	n	Class 0	Class 1	Entropy	Gini	Error	Weight
Parent	26,048	19,828	6,220	0.7930	0.3635	0.2388	1.0000
Child 1: age ≤ 30	8,514	7,974	540	0.3409	0.1188	0.0634	0.3269
Child 2: age > 30	17,534	11,854	5,680	0.9086	0.4380	0.3239	0.6731

The recorded weighted impurity values are: *H*_{weighted} = 0.7231, *G*_{weighted} = 0.3337, and *E*_{weighted} = 0.2388.

Calculated quantity	Numerical calculation	Result
Information gain	0.7930 – 0.7231	0.0700
Split information	$-\sum w_j \log_2(w_j)$, $w = (0.3269, 0.6731)$	0.9117
Gain ratio	0.0700 / 0.9117	0.0767
Gini reduction	0.3635 – 0.3337	0.0299
Misclassification-error reduction	0.2388 – 0.2388	≈ –0.0000

The worked example demonstrates that a split can show useful information gain and Gini reduction while providing essentially no reduction in the majority-class error. The different criteria therefore rank candidate splits differently because they emphasize different aspects of class purity.

6. Task 3 — Information-Gain Bias Experiment

To test the sensitivity of information gain to a high-cardinality, non-informative attribute, the experiment appends an artificial *UniqueID* feature containing a distinct identifier for every training example. The same setup is evaluated with entropy-based splitting and gain ratio, both at maximum depth 10.

Criterion	Root feature	Training accuracy	Validation accuracy
Entropy / Information Gain	<i>UniqueID</i>	100.00%	75.11%
Gain Ratio	<i>capital_gain</i>	87.01%	84.83%

Analysis: the UniqueID feature allows entropy-based splitting to memorize the training set, producing 100% training accuracy but only 75.11% validation accuracy. Gain ratio instead selects capital_gain and retains much stronger validation performance (84.83%). This experiment directly illustrates why normalizing information gain by split information can reduce the attraction of highly fragmented features.

7. Task 4 — Overfitting, Tree Complexity and Label Noise

The entropy criterion is evaluated at depth settings 2, 4, 6, 8, 10, 15, and unlimited. For each model, the notebook records training accuracy, validation accuracy, total nodes, leaves, and the actual maximum depth. The same experiment is then repeated after flipping 10% of the training labels at random.

7.1 Clean training data

Max-depth setting	Train acc.	Val. acc.	Total nodes	Leaves	Actual depth
2	82.98%	82.79%	47	40	3
4	85.82%	84.40%	924	750	5
6	89.17%	83.25%	2,822	2,008	7
8	92.11%	82.68%	4,392	2,916	9
10	94.36%	81.88%	5,616	3,573	11
15	97.70%	81.01%	7,477	4,526	16
Unlimited	100.00%	80.18%	8,879	5,229	45

7.2 10% label noise

Max-depth setting	Train acc.	Val. acc.	Total nodes	Leaves	Actual depth
2	76.34%	82.79%	116	109	3
4	79.29%	83.79%	1,497	1,244	5
6	83.73%	81.65%	4,267	3,129	7
8	88.26%	79.23%	6,810	4,591	9
10	91.82%	77.55%	8,827	5,648	11
15	96.76%	75.33%	11,771	7,159	16
Unlimited	99.98%	74.45%	13,712	8,139	44

Task 4: Overfitting & Noise Analysis (Criterion: ENTROPY)

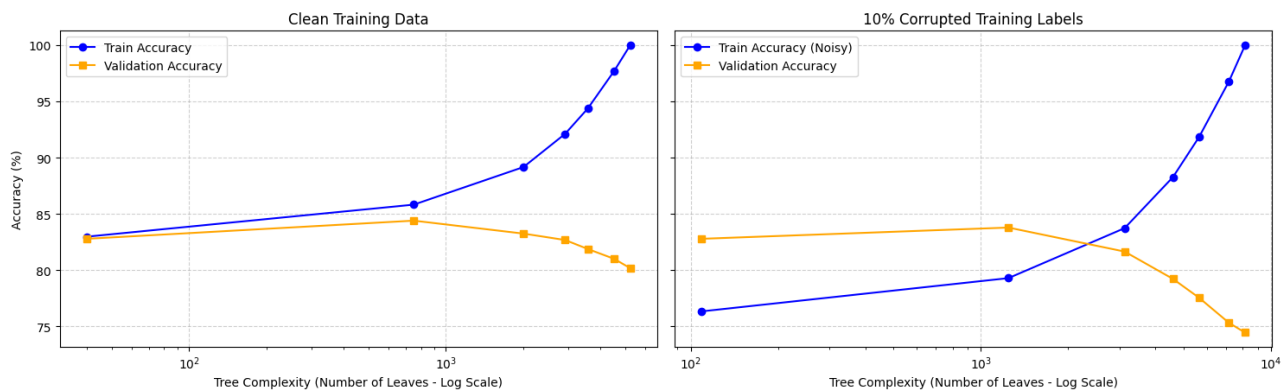


Figure 3. Task 4: training/validation accuracy versus number of leaves for clean data and 10% corrupted training labels.

Analysis: on clean data, training accuracy rises from 82.98% at depth 2 to essentially 100% for the unlimited tree, while validation accuracy peaks at 84.40% around depth 4 and then declines to 80.18%. The widening train–validation gap is a direct signature of overfitting. Under 10% label noise, validation accuracy peaks at 83.79% at depth 4 and declines much more sharply to 74.45% for the unlimited tree. At the same time, the noisy trees become substantially larger: the unlimited noisy model contains 13,712 nodes and 8,139 leaves compared with 8,879 nodes and 5,229 leaves on clean data.

8. Task 5 — Hyperparameter Selection and Final Evaluation

The notebook searches over three criteria—entropy, gain ratio, and Gini—and seven candidate maximum depths: 2, 4, 6, 8, 10, 12, and 15. The best configuration is selected by validation accuracy and then refit before the final evaluation call.

Selected criterion	Selected max depth	Best validation accuracy
Gain Ratio	15	85.17%

Metric	Recorded value
Accuracy	85.17%
Precision	0.7773
Recall	0.5663
F1-score	0.6552
True negatives	4,629
False positives	263
False negatives	703
True positives	918
Total nodes	848
Number of leaves	508
Actual maximum depth	16

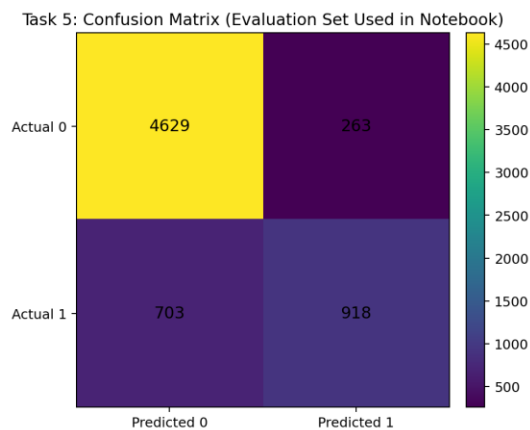


Figure 4. Confusion matrix reproduced from the counts reported by the notebook.

The confusion matrix contains 6,513 examples, matching the validation-set size. The comparatively high precision (0.7773) and lower recall (0.5663) indicate that the selected model is more conservative in assigning the positive-income class: when it predicts >50K it is usually correct, but it misses a substantial fraction of actual >50K cases.

9. Overall Analysis and Discussion

- Criterion behavior is data-dependent. Misclassification error is best in the fixed-depth Task 1 comparison, while gain ratio is selected by the broader Task 5 hyperparameter search.
- Categorical features can be extremely informative in this implementation. Relationship is the strongest root feature under both Gini reduction and information gain, producing a multiway six-category split.
- Gain ratio is useful when a feature has many possible partitions. The UniqueID experiment shows a sharp contrast: entropy selects the identifier and memorizes the training set, while gain ratio avoids it.
- Tree depth is a dominant complexity control. Validation accuracy peaks at a shallow tree and then declines as the number of leaves grows rapidly.
- Label noise amplifies overfitting. Deeper trees fit corrupted labels increasingly well while their validation performance deteriorates.
- The custom implementation directly supports mixed-type features and does not require one-hot encoding, which makes the decision rules interpretable in the original feature space.

9.1 Limitations and reproducibility notes

- The notebook uses a single 80/20 train-validation split, so the reported accuracies can depend on that split.
- The 10% label-noise experiment is randomized via `np.random.choice`; the global seed is set once at notebook start, but the noise experiment itself is a single draw rather than an averaged multi-run study.
- The outlier-removal section is commented out and therefore does not affect the reported experiments.
- The executed Task 5 call reuses validation data as the evaluation data. A separate untouched test set should be used for an unbiased final generalization estimate.
- The notebook contains an auxiliary `split()` helper whose `gain_ratio` call includes an extra `feature_values` argument, while `Metrics.gain_ratio` accepts only `parent_labels` and `child_splits`. The reported classifier experiments, however, use the `DecisionTreeClassifier` path and its `_calculate_score` method.

10. Team Contributions

Member	Contribution
Anshul Kaushal (2611AI10)	Contributed to the implementation and development of the decision-tree code and experiments.
Aditya Sharma (2611CS11)	Contributed to the implementation and development of the decision-tree code and experiments.
Roland Zohmingthana (2611MC03)	Contributed to preparation and development of the report.

The contribution summary reflects the allocation provided for this submission: Anshul and Aditya Sharma contributed to making the code, while Roland contributed to making the report.

11. Conclusion

The experiments provide a coherent empirical study of decision-tree split criteria and model complexity on the Adult income dataset. The implementation demonstrates how different impurity measures can produce different root splits and predictive performance, how gain ratio can reduce the attraction of high-cardinality identifiers, and how increasing depth trades generalization for training fit. Among the tested Task 5 settings, gain ratio with maximum depth 15 provides the best validation accuracy of 85.17% in the notebook. The strongest methodological caution is that this final evaluation is performed on the validation set because of the arguments used in the executed Task 5 call; a separate test set should be evaluated before using the result as a final generalization claim.

Appendix A — Experiment Checklist

Task	Experiment	Main output
1	Compare Gini, entropy, misclassification error, gain ratio at depth 5	Validation accuracy
2	Evaluate candidate root splits and manual age ≤ 30 calculation	Root feature, split score, impurity calculations
3	Inject UniqueID and compare entropy with gain ratio	Root feature + train/validation accuracy
4	Vary tree depth with clean and 10% noisy labels	Accuracy, nodes, leaves, actual depth
5	Search criterion $\times$ max depth	Selected configuration + final classification metrics